

AI Assistant for Smarter Learning & Study Support

Project Presentation

Problem Statement

Students spend significant time searching, reading and organizing study materials.

Manual summarization and question creation are time-consuming.

Students need personalized revision support and progress tracking.

Proposed Solution

LearnMate is an AI-powered study assistant.

Uploads study materials and processes learning content.

Generates summaries, flashcards, quizzes and study plans.

Provides secure APIs with role-based access.

Objectives

Reduce manual study effort.

Provide personalized learning assistance.

Generate useful revision content with AI.

Track materials, quizzes and study plans in one platform.

Technology Stack

Backend: Node.js + Express.js

Database: MongoDB + Mongoose

AI: Google Gemini API

Authentication: JWT + bcrypt

File Upload: Multer

API Testing: Postman

System Architecture

Client / Postman → Express REST API → Authentication & Controllers



MongoDB Database



Gemini AI Service



Summaries / Flashcards / Quizzes / Plans

Core Modules

Authentication & User Management

Study Material Upload

AI Summary Generation

Flashcard Generation

Quiz Generation

Personalized Study Plan

Admin / RBAC Management

ER / Data Model

User → owns → Material

Material → produces → Summary

Material → produces → Flashcard

Material → produces → Quiz

User → has → StudyPlan

Admin → manages → Users

MVC Architecture

Routes: define REST API endpoints

Controllers: handle application logic

Models: define MongoDB schemas

Middleware: authentication, authorization and uploads

Utils: Gemini AI integration and database configuration

Authentication & RBAC

User registration and login

Passwords protected using bcrypt hashing

JWT token used for authenticated requests

Student and Admin roles

Protected routes verify token and permissions

AI Processing Flow

1. Student uploads learning material
2. Backend validates and stores material
3. Gemini receives a learning-focused prompt
4. AI generates structured output
5. Backend stores the result in MongoDB
6. Student retrieves the result through REST APIs

REST API Endpoints

POST /api/auth/register

POST /api/auth/login

POST /api/materials

GET /api/materials

POST /api/ai/summary/:materialId

POST /api/ai/flashcards/:materialId

POST /api/ai/quiz/:materialId

POST /api/ai/study-plan

Sample Coding Structure

index.js

src/config/db.js

src/models/

src/controllers/

src/routes/

src/middleware/

src/utils/gemini.js

postman/

Project Setup

1. Install Node.js and MongoDB
2. Extract the project ZIP
3. Run: `npm install`
4. Create `.env` from `.env.example`
5. Add MongoDB URI, JWT secret and Gemini API key
6. Run: `npm run dev`

Postman Demo Flow

Register → Login → Copy JWT

Upload a material

Generate summary

Generate flashcards

Generate quiz

Create study plan

Verify protected endpoints

Scenario Outcome

Input: a student's learning material

Processing: backend + Gemini AI

Output: concise summary, revision flashcards and quiz

Result: a single workflow for faster study preparation

Future Enhancements

Real PDF/DOCX text extraction

Web/mobile frontend

Learning analytics dashboard

Spaced-repetition scheduling

Voice-based study assistance

More detailed progress recommendations

Conclusion

LearnMate combines REST APIs, MongoDB and Gemini AI into a practical study assistant.

It supports secure user access and AI-assisted learning workflows.

The architecture can be extended into a complete learning platform.

Thank You

LearnMate - AI Assistant for Smarter Learning & Study Support